Ex. No. 11	Create desktop applications using Java Swing
Date of Exercise	**30-03-2026**
Git-hub Host Link	https://github.com/chrisharrison-byte/Create-desktop-applications-using-Java-Swing

Aim

To develop a **desktop application using Java Swing** by designing a simple GUI for user interaction.

Description:

Java Swing is a part of Java Foundation Classes (JFC) used to build **Graphical User Interface (GUI)** applications.

Swing provides components such as:

- JFrame – Main window
- JLabel – Display text
- JTextField – Input field
- JButton – Perform actions

This experiment demonstrates:

- Creating a GUI window using JFrame
- Adding components like labels, text fields, and buttons
- Handling button click events using ActionListener
- Displaying output dynamically

Swing applications are widely used in **desktop tools, form-based applications, and management systems**.

Procedure:

Step 1 : Start the program.

Step 2 : Import the javax.swing and java.awt.event packages.

Step 3 : Create a class that extends JFrame.

Step 4 : Add GUI components such as JLabel, JTextField, and JButton.

Step 5 : Set layout and positions of components.

Step 6 : Add components to the frame.

Step 7 : Implement ActionListener to handle button click events.

Step 8 : Perform an action (e.g., display entered text) when the button is clicked.

Step 9 : Set frame properties like size and visibility.

Step 10 : Stop the program.

Program:

```
import javax.swing.*;
import java.awt.event.*;

public class SwingDemo extends JFrame implements ActionListener {

    JLabel label;
    JTextField textField;
    JButton button;

    SwingDemo() {
        // Create components
        label = new JLabel("Enter your name:");
        label.setBounds(50, 50, 150, 30);

        textField = new JTextField();
        textField.setBounds(200, 50, 150, 30);

        button = new JButton("Submit");
```

```
        button.setBounds(150, 100, 100, 30);

        // Add Action Listener
        button.addActionListener(this);

        // Add components to frame
        add(label);
        add(textField);
        add(button);

        // Frame settings
        setSize(400, 250);
        setLayout(null);
        setVisible(true);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    }

    public void actionPerformed(ActionEvent e) {
        String name = textField.getText();
        JOptionPane.showMessageDialog(this, "Hello " + name + "!");
    }

    public static void main(String[] args) {
        new SwingDemo();
    }
}
```

Output

Result

Thus, a **desktop application using Java Swing** was successfully created with GUI components and event handling.